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Lic. in Informatics and Computing Engineering (L.EIC)

Discrete Mathematics

5 - Proofs in First Order Logic

5.1. Indicate whether the following deductions are valid. If so, indicate the rules used in each step. If not valid, indicate which step it is not correct and if there is a correct deduction with the same premises and conclusion.

1	$\forall x(P(x) \vee Q(x))$	
2	u	$P(u) \vee Q(u)$
3		$P(u)$
4	$\forall x P(x)$	
5	$\forall x P(x) \vee \forall x Q(x)$	

1	$\forall x\,(P(x) \to Q(x))$	
2	$\exists x\,P(x)$	
3	v	$P(v)$
4		$P(v) \to Q(v)$
5		$Q(v)$
6	$\exists x\,Q(x)$	

1	$\exists x\,P(x)$	
2	u	$P(u)$
3	$\forall x\,P(x)$	

5.2. Using the natural deduction system for 1$^{\underline{\text{st}}}$ order logic and without using completeness, show using Fitch notation:

(a) $\exists x(R(x) \wedge S(x)) \vdash \exists x\,R(x) \wedge \exists x\,S(x)$
(b) $\forall x(R(x) \to S(x)) \vdash \forall x\,R(x) \to \forall x\,S(x)$
(c) $\forall x(P(x) \vee Q(x)), \exists x \neg Q(x), \forall x(R(x) \to \neg P(x)) \vdash \exists x \neg R(x)$
(d) $\forall x\,P(a, x, x), \forall x \forall y \forall z(P(x, y, z) \to P(f(x), y, f(z))) \vdash P(f(a), a, f(a))$
(e) $\vdash \forall x \forall y(\neg(f(x) = f(y)) \to \neg(x = y))$
(f) $\forall x(P(x) \to (Q(x) \vee R(x))), \neg\exists x(P(x) \wedge R(x)) \vdash \forall x(P(x) \to Q(x))$
(g) $\vdash \forall x \forall y \forall u \forall v((x = u \wedge y = v) \to f(x, y) = f(u, v)))$

5.3. For each of the following formulas, indicate whether or not it is a 1$^{\underline{\text{st}}}$ order logic theorem, whatever the formulas φ and ψ. Justify, if it is a theorem, showing that there is a natural deduction of the respective formula or, if it is not a theorem, indicating a language $\mathcal{L}$, formulas ψ and φ from $\mathcal{L}$ and a structure $\mathcal{A} = (A, \cdot^{\mathcal{A}})$ of $\mathcal{L}$, such that $\mathcal{A}$ is not a model of the corresponding formula.

(a) $\forall x(\varphi \to \psi) \to (\forall x\,\varphi \to \forall x\,\psi)$
$(\forall x\,\varphi \to \forall x\,\psi) \to \forall x(\varphi \to \psi)$

(b) $\forall x(\varphi \to \psi) \to (\exists x\ \varphi \to \exists x\ \psi)$
$(\exists x\ \varphi \to \forall x\ \psi) \to \forall x(\varphi \to \psi)$

(c) $\forall x \exists y(\varphi \vee \psi) \to \exists x \forall y(\varphi \wedge \psi)$
$\forall x(\psi \vee \varphi) \to \neg \exists x(\neg \psi \wedge \neg \varphi)$

5.4. Construct proofs for the following arguments:

(a) From $\forall x\, \forall y\, \forall z(\mathsf{R}(x, y) \wedge \mathsf{R}(y, z) \to \mathsf{R}(x, z))$ and $\forall x\, \neg \mathsf{R}(x, x)$ conclude $\forall x\, \forall y\, (\mathsf{R}(x, y) \to \neg \mathsf{R}(y, x))$;

(b) From $\forall x\, \forall y\, (\mathsf{R}(x, y) \to \neg \mathsf{R}(y, x))$ conclude $\forall x\, \neg \mathsf{R}(x, x)$;

(c) From $\forall x\, \forall y\, (\mathsf{R}(x, y) \wedge x \neq y \to \neg \mathsf{R}(y, x))$ conclude $\forall x\, \forall y\, (\mathsf{R}(x, y) \wedge \mathsf{R}(y, x) \to x = y)$.

	Introduction	**Elimination**
$\wedge$	φ $\vdots$ ψ $\vdots$ $\varphi \wedge \psi$ $\wedge$I	$\vdots$ $\varphi \wedge \psi$ $\vdots$ φ $\wedge$E $\vdots$ $\varphi \wedge \psi$ $\vdots$ ψ $\wedge$E
$\vee$	$\vdots$ φ $\vdots$ $\varphi \vee \psi$ $\vee$I $\vdots$ φ $\vdots$ $\psi \vee \varphi$ $\vee$I	$\varphi \vee \psi$ $\vdots$ φ / $\vdots$ / γ ψ / $\vdots$ / γ γ $\vee$E
$\neg$	φ / $\vdots$ / **F** $\neg\varphi$ $\neg$I	$\vdots$ $\neg\neg\varphi$ $\vdots$ φ $\neg$E
F	φ $\vdots$ $\neg\varphi$ $\vdots$ **F** **FI**	$\vdots$ **F** $\vdots$ φ **FE**
$\rightarrow$	φ / $\vdots$ / ψ $\varphi \rightarrow \psi$ $\rightarrow$I	φ $\vdots$ $\varphi \rightarrow \psi$ $\vdots$ ψ $\rightarrow$E
$=$	$\vdots$ $t = t$ $=_{\mathbf{I}}$	$t_1 = t_2$ $\vdots$ $\varphi[x \mapsto t_1]$ $\vdots$ $\varphi[x \mapsto t_2]$ $=_{\mathbf{E}}$ and x is replaceable by t_1 and by t_2 in φ
$\forall$	ν / $\vdots$ / $\varphi[x \mapsto \nu]$ $\forall x\ \varphi$ $\forall_{\mathbf{I}}$ where ν is a new variable (does not occur before)	$\forall x\ \varphi$ $\vdots$ $\varphi[x \mapsto t]$ $\forall_{\mathbf{E}}$ where x is replaceable by t in φ
$\exists$	$\varphi[x \mapsto t]$ $\vdots$ $\exists x\ \varphi$ $\exists_{\mathbf{I}}$ where x is replaceable by t in φ	$\exists x\ \varphi$ ν / $\varphi[x \mapsto \nu]$ / $\vdots$ / ψ ψ $\exists_{\mathbf{E}}$ where ν is a new variable that does not occur before or in ψ